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Extended State Observer Based Robust Nonlinear PID Attitude Tracking Control of Quadrotor with Lumped Disturbance

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Abstract: The paper presents a robust nonlinear PID controller for the attitude tracking problem of quadrotors subject to disturbance. First, to suppress the influence caused by external disturbance torque, considering the fact that the angular velocity can be obtained by the inertial measurement unit (IMU), a reduced-order extended state observer (ESO) is applied as a feedforward compensation to improve the robustness of the tracking system. Then, an ESO-based nonlinear PID controller is constructed to track the desired attitude command, and the rigorous proof of the convergence of the closed-loop system is derived by utilizing the Lyapunov method. Finally, the effectiveness of the proposed method is illustrated by numerical simulations and platform experiments.

Keywords: quadrotor; nonlinear PID control; disturbance rejection; real-time experiments

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Academic Editors: Yuanqiang Zhou, Le Yao, Xiaoyu Jiang and Zheren Zhu

Received: 15 March 2025
Revised: 5 May 2025
Accepted: 7 May 2025
Published: 12 May 2025

Citation: Xu, G.; Luo, S.; Huang, Y.; Deng, X. Extended State Observer Based Robust Nonlinear PID Attitude Tracking Control of Quadrotor with Lumped Disturbance. *Processes* **2025**, *13*, 1470. https://doi.org/10.3390/pr13051470

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1. Introduction

Quadrotor unmanned aerial vehicles (UAVs) have become increasingly vital across a spectrum of applications, including aerial surveillance, inspection, precision agriculture, and disaster response, owing to their vertical take-off and landing (VTOL) capabilities, mechanical simplicity, and high maneuverability [1–3]. However, their practical deployment remains constrained by inherent challenges such as strong nonlinear dynamics, underactuation, and high sensitivity to environmental disturbances like wind gusts, payload variations, and sensor noise. These limitations underscore the critical importance of robust and precise attitude control for achieving stable and reliable flight performance.

Traditionally, PID controllers have been the choice for attitude stabilization due to their simplicity and ease of implementation [4]. In [5], the PID algorithm-based flight control was constructed for a quadrotor to study the dynamic control performance under different wind scenarios. To improve the quadrotor control performance, the feedback linearization combined with PID was proposed in [6]. However, in view of disadvantages of linear PID as mentioned in [7], the exploitation of nonlinear PID has further gained interest of researchers [8–12]. In [8], by using error scaling in the integral operation, the nonlinear PID controller for attitude stabilization was developed. To address time and state constraint problem of a quadrotor, a finite-time stability scheme with implicit PID controller was developed in [9]. In [10], a nonlinear PID controller was proposed for a quadrotor to track the given trajectory with minimum energy, and the parameters were tuned by genetic algorithm. In [11], motion control of a quadrotor was achieved by utilizing the designed nonlinear PID-type controller; the rigorous theoretical analysis of the closed-loop system and gain tuning guidelines were provided. Nevertheless, one persistent shortcoming across both linear and nonlinear PID approaches is their passive disturbance rejection. While some

robustness is inherently built into controller structures, explicit disturbance estimation and compensation remain underexplored. This is particularly critical given that real-world disturbances can be nontrivial and unpredictable.

To address this, recent attention has shifted towards disturbance observer (DOB) and extended state observer (ESO) [13–16], which actively estimate and compensate for lumped disturbances—comprising model uncertainties, sensor noise, and environmental forces. DOBs have been shown to enhance tracking under bounded disturbances [17], while ESO-based designs enable real-time estimation of both internal and external uncertainties [18–24]. For example, in [13], variable DOB control strategy was presented to improve disturbance estimation accuracy. By considering external disturbances, parametric uncertainties, input delay, and actuator fault as lumped disturbances, [14] proposed a fixed-time DOB based trajectory control scheme for a quadrotor. In [15], the integral sliding mode based ESO was developed for a quadrotor in the presence of actuator faults, which ensures the estimation errors accurately converge to the origin. In order to deal with unknown external disturbance, the composite nonlinear ESO, combining the advantages of linear and nonlinear ESOs, was proposed in [16] for trajectory tracking control of a quadrotor. In [18], adaptive ESO based prescribed-time attitude stabilization of a quadrotor was studied, in which ESO was used to estimate and compensate for complicated disturbed factors. However, most existing designs are full-order observers, which introduce undesirable phase lag and are sensitive to high-frequency noise, especially in high-bandwidth systems such as UAVs [25]. Additionally, few studies rigorously address the simultaneous compensation of lumped disturbances with reduced-order ESO within a nonlinear PID framework despite their potential to reduce computational burden and enhance active disturbance rejection while preserving design simplicity and improving real-time applicability [26]. Furthermore, experimental validation of ESO-based control strategies under realistic, noisy, and dynamic environments remains sparse [27], often confined to simulations or disturbance-free setups.

Motivated by the preceding observations, this paper aims to address these issues by proposing a reduced-order ESO-based nonlinear PID controller for quadrotor attitude tracking and bridging the gap between theoretical disturbance compensation techniques and their practical deployment in real-world aerial robotics. Main contributions include: (1) proposing a reduced-order ESO that leverages directly measurable angular velocities to estimate lumped disturbances with reduced phase lag and noise sensitivity; (2) designing a robust nonlinear PID controller that actively compensates for these disturbances in real time and providing rigorous Lyapunov-based stability proofs for the proposed closed-loop system; (3) demonstrating experimental validation on a quadrotor platform, including scenarios with payload-induced disturbances and sensor noise.

The remainder of this paper is organized as follows: Section 2 presents the quadrotor attitude dynamics and some useful properties. In Section 3, the nonlinear PID controller with disturbance rejection based on ESO is developed. Section 4 provides simulation and experimental results to show the control performance and the robustness against disturbance. Finally, conclusions are drawn in Section 5.

Notations: In this paper, the following notations are adopted: $\lambda_m(A)$ and $\lambda_M(A)$ stand for the minimum and maximum eigenvalue of a symmetric matrix $A \in \mathbb{R}^{3\times 3}$, respectively; $\|x\| = \sqrt{x^T x}$ is the Euclidean norm of vector $x \in \mathbb{R}^3$; and $\|A\| = \sqrt{\lambda_M(A^T A)}$ is the induced norm of matrix $A \in \mathbb{R}^{3\times 3}$.

2. Quadrotor Attitude Dynamics

2.1. Attitude Dynamics

In this paper, the quadrotor attitude dynamics are modeled as a fully actuated rigid body. As shown in Figure 1, for the convenience of establishing a mathematical model for

a quadrotor, inertial reference frame and body-fixed frame are defined as $\mathcal{I} = [x_I, y_I, z_I]$ and $\mathcal{B} = [x_B, y_B, z_B]$, respectively. The orientation of the body-fixed frame with respect to the inertial frame is denoted by $\eta = [\phi, \theta, \psi]^T \in \mathbb{R}^3$, composed of the three Euler angles. These angles are bounded as follows: roll angle, ϕ, by $(-\pi/2 < \phi < \pi/2)$; pitch angle, θ, by $(-\pi/2 < \theta < \pi/2)$; yaw angle, ψ, by $(-\pi < \psi < \pi)$.

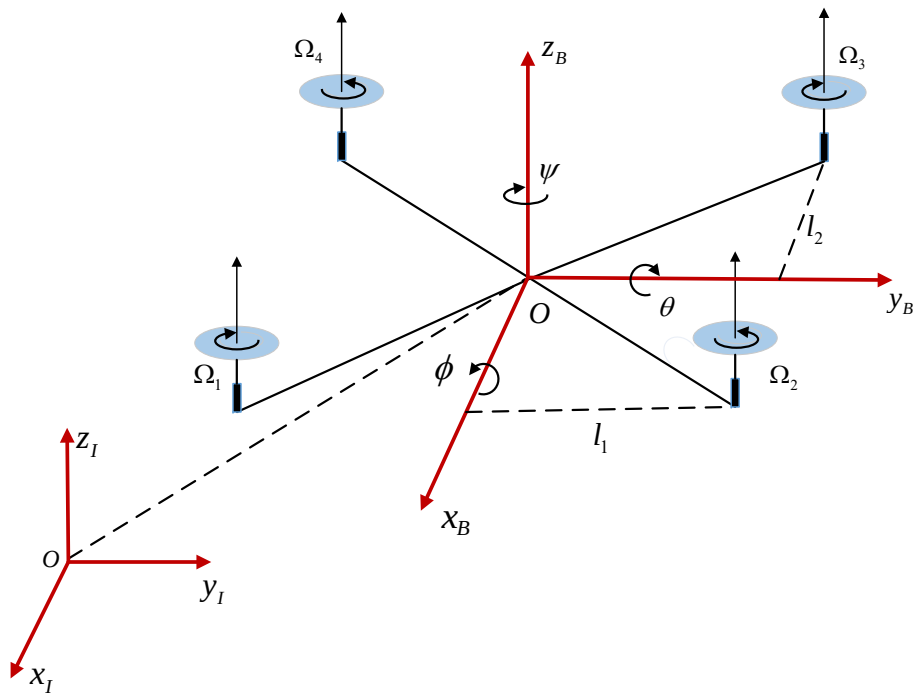


Figure 1. Body-fixed frame and earth-fixed frame for the quadrotor.

The quadrotor attitude dynamic can be expressed by the Lagrange–Euler equation as follows [28]:

$$M(\eta)\ddot{\eta} + C(\eta, \dot{\eta})\dot{\eta} = \tau + \tau_d \tag{1}$$

where $M(\eta) \in \mathbb{R}^{3\times3}$ denotes the inertia matrix, $C(\eta, \dot{\eta}) \in \mathbb{R}^{3\times3}$ represents the centripetal–Coriolis matrix, $\tau \in \mathbb{R}^3$ and $\tau_d \in \mathbb{R}^3$ are the control torque and external disturbance torque, represented with respect to the body-fixed frame, respectively.

Matrices $M(\eta)$ and $C(\eta, \dot{\eta})$ are in the following form:

$$M(\eta) = \begin{bmatrix} I_{xx} & 0 & -I_{xx}S_\theta \\ 0 & I_{yy}C_\phi^2 + I_{zz}S_\phi^2 & (I_{yy} - I_{zz})C_\phi S_\phi C_\theta \\ -I_{xx}S_\theta & (I_{yy} - I_{zz})C_\phi S_\phi C_\theta & I_{xx}S_\theta^2 + I_{yy}S_\phi^2 C_\theta^2 + I_{zz}C_\phi^2 C_\theta^2 \end{bmatrix} \tag{2}$$

and

$$C(\eta, \dot{\eta}) = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \tag{3}$$

with

$$
\begin{aligned}
c_{11} &= 0 \\
c_{12} &= (I_{yy} - I_{zz})\left(\dot{\theta}C_\phi S_\phi + \dot{\psi}S_\phi^2 C_\theta - \dot{\psi}C_\phi^2 C_\theta\right) - I_{xx}\dot{\psi}C_\theta \\
c_{13} &= (I_{zz} - I_{yy})\dot{\psi}C_\phi S_\phi C_\theta^2 \\
c_{21} &= (I_{zz} - I_{yy})\left(\dot{\theta}C_\phi S_\phi + \dot{\psi}S_\phi^2 C_\theta - \dot{\psi}C_\phi^2 C_\theta\right) + I_{xx}\dot{\psi}C_\theta \\
c_{22} &= (I_{zz} - I_{yy})\dot{\phi}C_\phi S_\phi \\
c_{23} &= -I_{xx}\dot{\psi}S_\theta C_\theta + I_{yy}\dot{\psi}S_\phi^2 C_\theta S_\theta + I_{zz}\dot{\psi}C_\phi^2 S_\theta C_\theta \\
c_{31} &= (I_{yy} - I_{zz})\dot{\psi}C_\phi S_\phi C_\theta^2 - I_{xx}\dot{\theta}C_\theta \\
c_{32} &= (I_{zz} - I_{yy})\left(\dot{\theta}C_\phi S_\phi S_\theta + \dot{\phi}S_\phi^2 C_\theta\right) + (I_{yy} - Izz)\dot{\phi}C_\phi^2 C_\theta \\
&\quad + I_{xx}\dot{\psi}S_\theta C_\theta - I_{yy}\dot{\psi}S_\phi^2 C_\theta S_\theta - I_{zz}\dot{\psi}C_\phi^2 S_\theta C_\theta \\
c_{33} &= (I_{yy} - I_{zz})\dot{\phi}C_\phi S_\phi C_\theta^2 - I_{yy}\dot{\theta}S_\phi^2 C_\theta S_\theta - I_{zz}\dot{\theta}C_\phi^2 C_\theta S_\theta \\
&\quad + I_{xx}\dot{\theta}C_\theta S_\theta
\end{aligned} \tag{4}
$$

where $S. = \sin(\cdot)$, $C. = \cos(\cdot)$, I_{xx}, I_{yy}, I_{zz} are moments of inertia around x, y, z axis, respectively.

The control torque τ is calculated as

$$
\tau = \begin{bmatrix} -k_T l_1 & k_T l_1 & k_T l_1 & -k_T l_1 \\ k_T l_2 & k_T l_2 & -k_T l_2 & -k_T l_2 \\ k_\tau & -k_\tau & k_\tau & -k_\tau \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \tag{5}
$$

where k_T is the thrust factor, k_τ is the drag factor, ω_i is the angular velocity of the ith motor ($i = 1, 2, 3, 4$), and l_1 and l_2 represent the distances from the rotors to the x_B and y_B axes, respectively.

Considering the boundaries of the system, the following assumptions are presumed for the following controller design:

Assumption 1. *The desired trajectory $\eta_d(t) \in \mathbb{R}^3$ satisfies $\eta_d(t), \dot{\eta}_d(t), \ddot{\eta}_d(t), \dddot{\eta}_d(t) \in \mathcal{L}_\infty$, and $\|\dot{\eta}_d\| \leq \nu$.*

Assumption 2. *The nonlinear disturbance term and its first time derivative are bounded, i.e., $\tau_d(t), \dot{\tau}_d(t) \in \mathcal{L}_\infty$.*

Remark 1. *Assumption 1 implies that there is no chance to produce an infinite control torque. Assumptions 1 and 2 ensure that the lumped disturbance mentioned later is bounded and differentiable.*

2.2. Useful Properties

Recalling the attitude dynamics in (1), the following properties for nominal dynamics are established:

Property 1. *The matrix $M(\eta)$ is symmetric, positive definite, and bounded by*

$$
0 < \lambda_m(M) \leq \|M(\eta)\| \leq \lambda_M(M) \tag{6}
$$

Property 2. *The matrix $\dot{M}(\eta) - 2C(\eta, \dot{\eta})$ is skew-symmetric, i.e.,*

$$\varsigma^T \left(\dot{M}(\eta) - 2C(\eta, \dot{\eta})\right)\varsigma = 0, \quad \forall \eta, \dot{\eta}, \varsigma \in \mathbb{R}^3 \tag{7}$$

Property 3. *The matrices $M(\eta)$ and $C(\eta, \dot{\eta})$ satisfy*

$$\dot{M}(\eta) = C(\eta, \dot{\eta}) + C(\eta, \dot{\eta})^T \tag{8}$$

Property 4. *There exists a positive constant C_M such that*

$$\|C(\eta, \dot{\eta})\| \leq C_M \|\dot{\eta}\| \tag{9}$$

3. Controller Design

3.1. Objective

The control objective is to develop a controller that will enable the attitude dynamics (1) to track a desired smooth trajectory η_d. To quantify the objective, an attitude tracking error is defined as $e_\eta(t) = \eta - \eta_d \in \mathbb{R}^3$. Then, the error dynamics can be written as follows:

$$M(\eta)\ddot{e}_\eta = -C(\eta, \dot{\eta})\dot{e}_\eta + \tau + f \tag{10}$$

where $f = -M(\eta)\ddot{\eta}_d - C(\eta, \dot{\eta})\dot{\eta}_d + \tau_d$ is considered as the lumped disturbance. To simplify the subsequent controller design, a portion of desired dynamics is also involved.

3.2. Disturbance Estimation

In this paper, to improve the robustness of the attitude tracking system, ESO is applied for a feedforward compensation. Since the angular velocity can be measured by IMU practically, a reduced-order ESO is considered here to reduce the phase lag and the influence of measurement noises. We define $x_1 = \dot{e}_\eta$, $x_2 = M^{-1}f$. According to Assumptions 1 and 2 and (6) in Property 1, $h(t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t}\left(M^{-1}f\right)$ is bounded. Then attitude error dynamics (10) can be rewritten as follows:

$$\dot{x}_1 = x_2 + f_0 + u \tag{11}$$

$$\dot{x}_2 = h(t) \tag{12}$$

where $f_0 = -M^{-1}C\dot{e}_\eta$ and $u = M^{-1}\tau$.

Then, the dynamics can be regarded as decoupled systems, and it has

$$\dot{x}_{1i} = x_{2i} + f_{0i} + u_i \tag{13}$$

$$\dot{x}_{2i} = h_i \tag{14}$$

where x_i denotes the ith element of vector $x \in \mathbb{R}^3$, and $i = 1, 2, 3$ stands for roll, pitch, and yaw, respectively.

For system (13), the ESO is usually in the following form:

$$\dot{z}_{1i} = z_{2i} + g_1\left(\frac{x_{1i} - z_{1i}}{\varepsilon}\right) + f_{0i} + u_i \tag{15}$$

$$\dot{z}_{2i} = \frac{1}{\varepsilon}g_2\left(\frac{x_{1i} - z_{1i}}{\varepsilon}\right) \tag{16}$$

where ε is a small adjustable parameter, and $g_1,\ g_2 : \mathbb{R} \to \mathbb{R}$ are selectable linear or nonlinear functions.

The following assumption is made for the ESO:

Assumption 3. *There exist constants λ_1, λ_2, λ_3, λ_4, α, and β and positive definite, continuous differentiable functions V_1, $W_1 : \mathbb{R}^2 \to \mathbb{R}$ such that for $y = [y_1, y_2]^T$*

$$\begin{aligned}
&\lambda_1\|y\|^2 \le V_1(y) \le \lambda_2\|y\|^2 \\
&\lambda_3\|y\|^2 \le W_1(y) \le \lambda_4\|y\|^2, \\
&\frac{\partial V_1}{\partial y_1}(y_2 - g_1(y_1)) - \frac{\partial V_1}{\partial y_2} g_2(y_1) \le -W_1(y) \\
&\left|\frac{\partial V_1}{\partial y_2}\right| \le \beta\|y\|
\end{aligned} \tag{17}$$

In order to illustrate the convergence of the presented ESO, and the following lemma is useful:

Lemma 1 ([17])**.** *Suppose that Assumptions 1–3 are satisfied; then, for ESO (15) and (16)*

$$\lim_{\varepsilon\to 0} |e_{ji}| = 0, \quad t \in [T_1, \infty) \tag{18}$$

$$\lim_{t\to\infty} \frac{|e_{ji}|}{\varepsilon^{3-j}} \le \varpi_{0i} \tag{19}$$

where $e_{ji} = z_{ji} - x_{ji}$, $j = 1, 2$, T_1 is a constant independent of ε, $\varpi_{0i} = \beta\bar{h}\lambda_2/(\lambda_1\lambda_3)$ is a positive constant.

According to the ESO (15) and (16), a nonlinear ESO is constructed for system (13) by choosing the functions as follows:

$$g_1(y) = 2\omega y_1 + \beta \mathrm{fal}(y_1, \varsigma, \delta) \tag{20}$$

$$g_2(y) = \omega^2 y_1 \tag{21}$$

where $0 < \varsigma < 1$, ω, β, δ are positive constants, and

$$\mathrm{fal}(y_1, \varsigma, \delta) = \begin{cases} y_1/\delta^{1-\varsigma} & |y_1| < \delta \\ |y_1|^{\varsigma}\,\mathrm{sign}(y_1) & |y_1| \ge \delta \end{cases} \tag{22}$$

For simplicity, $\mathrm{fal}(y_1, \varsigma, \delta)$ is abbreviated to $\mathrm{fal}(y_1)$. Then, the ESO is obtained as follows:

$$\dot{z}_{1i} = z_{2i} - \frac{2\omega}{\varepsilon} e_{1i} - \beta \mathrm{fal}\left(\frac{e_{1i}}{\varepsilon}\right) + f_{0i} + u_i \tag{23}$$

$$\dot{z}_{2i} = -\frac{\omega^2}{\varepsilon^2} e_{1i} \tag{24}$$

Theorem 1. *Suppose that Assumptions 1 and 2 are satisfied, and the observer is parameterized as*

$$\beta \le \frac{5\omega^2 + 1}{2\omega} \delta^{1-\varsigma} \tag{25}$$

then, for ESO (23) and (24),

$$\lim_{t\to\infty} \frac{|e_{ji}|}{\varepsilon^{3-j}} \le \varpi_i \tag{26}$$

where $j = 1, 2$ and ϖ_i is a positive constant.

Proof. Consider the Lyapunov function $V_1 : \mathbb{R}^2 \to \mathbb{R}$ given by

$$V_1(y) = y^T P y + \beta \int_0^{y_1} \mathrm{fal}(s)\mathrm{d}s \tag{27}$$

where P is a positive definite matrix that is the solution of a Lyapunov equation $A^T P + PA = -I$, and $I \in \mathbb{R}^{2\times 2}$ is the identity matrix. For ESO (23) and (24), A is obtained as

$$A = \begin{bmatrix} -2\omega & 1 \\ -\omega^2 & 0 \end{bmatrix}$$

and thus

$$P = \begin{bmatrix} \frac{1+\omega^2}{4\omega} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5\omega^2+1}{4\omega^3} \end{bmatrix}$$

Now we will prove that all conditions of Assumption 3 are satisfied. First, denote $\mu = \sqrt{\omega^2 + 6\omega^2 + 1}$ and then $\lambda_m(P) = \frac{\mu^2 - \mu\omega^2 - \mu}{8\omega^3}$, $\lambda_M(P) = \frac{\mu^2 + \mu\omega^2 + \mu}{8\omega^3}$. Hence,

$$\lambda_m(P)\|y\|^2 \le y^T P y \le \lambda_M(P)\|y\|^2 \tag{28}$$

Consider the term $\int_0^{y_1} \mathrm{fal}(s)\mathrm{d}s$ in two cases. If $|y_1| < \delta$, then

$$\int_0^{y_1} \mathrm{fal}(s)\mathrm{d}s = \int_0^{y_1} \delta^{\varsigma-1} s \,\mathrm{d}s = \frac{1}{2}\delta^{\varsigma-1} y_1^2 \le \frac{1}{2}\delta^{\varsigma-1}\|y\|^2 \tag{29}$$

If $|y_1| \ge \delta$, then

$$\int_0^{y_1} \mathrm{fal}(s)\mathrm{d}s \le \int_0^{y_1} \delta^{\varsigma-1} s \,\mathrm{d}s = \frac{1}{2}\delta^{\varsigma-1} y_1^2 \le \frac{1}{2}\delta^{\varsigma-1}\|y\|^2 \tag{30}$$

It is obvious that $\int_0^{y_1} \mathrm{fal}(s)\mathrm{d}s \ge 0$. From (28) to (30), it is obtained that

$$\lambda_m(P)\|y\|^2 \le V_1(y) \le \left(\lambda_M(P) + \frac{1}{2}\delta^{\varsigma-1}\right)\|y\|^2 \tag{31}$$

Then, a direct computation shows that

$$\begin{aligned} &\frac{\partial V_1}{\partial y_1}(y_2 - g_1(y_1)) - \frac{\partial V_1}{\partial y_2} g_2(y_1) \\ =\ & y_1^2 - y_2^2 - \beta^2 \mathrm{fal}^2(y_1) - \frac{5\omega^2+1}{2\omega} y_1 \mathrm{fal}(y_1) \\ & + 2\beta y_2 \mathrm{fal}(y_1) \end{aligned} \tag{32}$$

Invoking Young's inequality, it has

$$2\beta y_2 \mathrm{fal}(y_1) \le \frac{1}{2} y_2^2 + 2\beta^2 \mathrm{fal}^2(y_1) \tag{33}$$

Then

$$\begin{aligned} &\frac{\partial V_1}{\partial y_1}(y_2 - g_1(y_1)) - \frac{\partial V_1}{\partial y_2} g_2(y_1) \\ \le\ & -y_1^2 - \frac{1}{2} y_2^2 + \beta^2 \mathrm{fal}^2(y_1) - \frac{5\omega^2+1}{2\omega} y_1 \mathrm{fal}(y_1) \\ =\ & -y_1^2 - \frac{1}{2} y_2^2 - \beta \mathrm{fal}(y_1)\left(\frac{5\omega^2+1}{2\omega} y_1 - \beta \mathrm{fal}(y_1)\right) \end{aligned} \tag{34}$$

Note that $\mathrm{fal}(y_1)(\frac{5\omega^2+1}{2\omega}y_1 - \beta\mathrm{fal}(y_1)) \geq 0$ always holds when (25) is satisfied, and thus

$$\begin{aligned} &\frac{\partial V_1}{\partial y_1}(y_2 - g_1(y_1)) - \frac{\partial V_1}{\partial y_2}g_2(y_1) \\ \leq\ & -\omega^2 y_1^2 - \frac{1}{2}y_2^2 \triangleq -W_1(y) \end{aligned} \tag{35}$$

It is obvious that

$$\min\left\{1, \frac{1}{2}\right\}\|y\|^2 \leq W_1(y) \leq \max\left\{1, \frac{1}{2}\right\}\|y\|^2 \tag{36}$$

and

$$\left|\frac{\partial V_1}{\partial y_2}\right| = \left|-y_1 + \frac{5\omega^2+1}{2\omega^3}y_2\right| \leq \max\left\{1, \frac{5\omega^2+1}{2\omega^3}\right\}\|y\| \tag{37}$$

Equations (31)–(37) indicate that all conditions of Assumption 3 are satisfied. Therefore, the convergence of ESO (23)–(24) is guaranteed according to Lemma 1. The proof is completed. □

From the discussion above, the estimation error of ESO is bounded. We define the estimation error of disturbance f as $e_f = Mz_2 - f$, and thus e_f is also bounded, i.e.,

$$\|e_f\| = \|Mz_2 - f\| \leq \bar{e}_f \tag{38}$$

3.3. Attitude Tracking Control

Here, the problem of quadrotor dynamics tracking the desired attitude command η_d is considered. To aid the subsequent controller design and stability analysis, the vector $\mathrm{Atan}(\cdot) \in \mathbb{R}^3$ and the diagonal matrix $T(\cdot) \in \mathbb{R}^{3\times 3}$ are defined as follows:

$$\mathrm{Atan}(x) = [\mathrm{atan}(x_1)\ \mathrm{atan}(x_2)\ \mathrm{atan}(x_3)]^T \tag{39}$$

$$T(x) = \mathrm{diag}\left\{\frac{1}{1+x_i^2}\right\},\ i = 1,2,3 \tag{40}$$

for $x = [x_1\ x_2\ x_3]^T \in \mathbb{R}^3$. It is obvious that the following expressions hold:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Atan}(x) = T(x)\dot{x} \tag{41}$$

$$\|\mathrm{Atan}(x)\| \leq \frac{\sqrt{3}}{2}\pi \tag{42}$$

$$\|T(x)\| \leq 1 \tag{43}$$

Based on the control objective and the subsequent stability analysis, the proposed nonlinear PID controller is formulated as follows:

$$\tau = -K_p\mathrm{Atan}(\sigma e_\eta) - K_i\mathrm{Atan}(\xi) - K_d\dot{e}_\eta - Mz_2 \tag{44}$$

where $K_p, K_i, K_d \in \mathbb{R}^{3\times 3}$ are diagonal positive definite matrices, and σ is a positive constant; $\xi \in \mathbb{R}^3$ is given by the filter

$$\dot{\xi} = T(\xi)^{-1}\Big(-K_f\mathrm{Atan}(\xi) + \gamma\mathrm{Atan}(\sigma e_\eta) + \rho\dot{e}_\eta\Big) \tag{45}$$

where $K_f \in \mathbb{R}^{3\times 3}$ is a diagonal positive definite matrix, and γ and ρ are positive constants.

Theorem 2. *For system (1) controlled by (44), if the controller is parameterized as*

$$\sigma \leq \min\{\sigma_1, \sigma_2\} \tag{46}$$

where

$$\begin{aligned}
\sigma_1 &= \frac{\rho\lambda_m(M)\Big(\rho\lambda_m(K_p) + \gamma\lambda_m(K_d)\Big)}{\gamma^2\lambda_M^2(M)} \\
\sigma_2 &= \frac{\lambda_m(K_p)\Big(\rho\lambda_m(K_d) - \frac{\sqrt{3}}{2}\pi\gamma C_M\Big) - \frac{\gamma}{4}C_M^2\nu^2}{\gamma\lambda_m(K_p)\lambda_M(M)}
\end{aligned}$$

then the closed-loop system is uniformly ultimately bounded (UUB).

Proof. Consider the following Lyapunov function:

$$\begin{aligned}
V_2 &= \frac{1}{2}\rho\dot{e}_\eta^T M\dot{e}_\eta + \gamma \mathrm{Atan}(\sigma e_\eta)^T M\dot{e}_\eta \\
&\quad + \rho\sigma^{-1}\sum_{i=1}^{3} k_{pi}\Big(\sigma e_{\eta i}\mathrm{atan}(\sigma e_{\eta i}) - \frac{1}{2}\ln(1+\sigma^2 e_{\eta i}^2)\Big) \\
&\quad + \gamma\sigma^{-1}\sum_{i=1}^{3} k_{di}\Big(\sigma e_{\eta i}\mathrm{atan}(\sigma e_{\eta i}) - \frac{1}{2}\ln(1+\sigma^2 e_{\eta i}^2)\Big) \\
&\quad + \frac{1}{2}\mathrm{Atan}(\xi)^T K_i \mathrm{Atan}(\xi)
\end{aligned} \tag{47}$$

where $k_{pi}, k_{di} \in \mathbb{R}$ are the ith diagonal entries of K_p and K_d, respectively, and $e_{\eta i} \in \mathbb{R}$ is the ith element of e_η.

First, the range of parameters is to be determined to guarantee positive definiteness of V_2. Using the fact

$$\sum_{i=1}^{3}\left[x_i\mathrm{atan}(x_i) - \frac{1}{2}\ln(1+x_i^2)\right] \geq \frac{1}{2}\|\mathrm{Atan}(x)\|^2$$

for $x = [x_1\ x_2\ x_3]^T \in \mathbb{R}^3$, yields

$$\begin{aligned}
V_2 &\geq \frac{1}{2}\rho\lambda_m(M)\|\dot{e}_\eta\|^2 - \gamma\lambda_M(M)\|\mathrm{atan}(\sigma e_\eta)\|\|\dot{e}_\eta\| \\
&\quad + \frac{1}{2\sigma}\Big(\rho\lambda_m(K_p) + \gamma\lambda_m(K_d)\Big)\|\mathrm{atan}(\sigma e_\eta)\|^2
\end{aligned} \tag{48}$$

Applying Young's inequality, there is

$$\begin{aligned}
V_2 &\geq \left(\sqrt{\sigma^{-1}\rho\lambda_m(M)\big(\rho\lambda_m(K_p) + \gamma\lambda_m(K_d)\big)} - \gamma\lambda_M(M)\right) \\
&\quad \|\mathrm{atan}(\sigma e_\eta)\|\|\dot{e}_\eta\|
\end{aligned} \tag{49}$$

and thus V_2 is positive definite if

$$\sigma \leq \frac{\rho\lambda_m(M)\Big(\rho\lambda_m(K_p) + \gamma\lambda_m(K_d)\Big)}{\gamma^2\lambda_M^2(M)} \tag{50}$$

which is always satisfied if (46) holds.

Then the negativity of $\dot{V}_2$ is to be considered. Substituting (44) into (10) yields

$$M\ddot{e}_\eta = -C\dot{e}_\eta - K_p\mathrm{Atan}(\sigma e_\eta) - K_i\mathrm{Atan}(\xi) - K_d\dot{e}_\eta - e_f \tag{51}$$

Using the fact

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{3} k_{pi}\left[\sigma e_{\eta i}\mathrm{atan}(\sigma e_{\eta i}) - \frac{1}{2}\ln(1+\sigma^2 e_{\eta i}^2)\right] \\ =\ &\sigma\dot{e}_\eta^T K_p\mathrm{Atan}(\sigma e_\eta) \end{aligned} \tag{52}$$

then the derivative of V_2 is calculated as follows:

$$\begin{aligned} \dot{V}_2 =\ & \frac{1}{2}\rho\dot{e}_\eta^T\dot{M}\dot{e}_\eta + \rho\dot{e}_\eta^T M\ddot{e}_\eta + \gamma\sigma\dot{e}_\eta^T T(\sigma e_\eta)M\dot{e}_\eta \\ & +\gamma\mathrm{Atan}(\sigma e_\eta)^T\dot{M}\dot{e}_\eta + \gamma\mathrm{Atan}(\sigma e_\eta)^T M\ddot{e}_\eta \\ & +\rho\dot{e}_\eta^T K_p\mathrm{Atan}(\sigma e_\eta) + \gamma\dot{e}_\eta^T K_d\mathrm{Atan}(\sigma e_\eta) \\ & +\mathrm{Atan}(\xi)^T K_i T(\xi)\dot{\xi} \end{aligned} \tag{53}$$

Substituting (44), (51) into (53), and recalling (7) in Property 2 and (8) in Property 3, yields

$$\begin{aligned} \dot{V}_2 =& -\rho\dot{e}_\eta^T K_d\dot{e}_\eta - \rho\dot{e}_\eta^T e_f + \gamma\sigma\dot{e}_\eta^T T(\sigma e_\eta)M\dot{e}_\eta \\ & +\gamma\mathrm{Atan}(\sigma e_\eta)^T C(\eta,\dot{\eta})^T\dot{e}_\eta - \gamma\mathrm{Atan}(\sigma e_\eta)^T K_p\mathrm{Atan}(\sigma e_\eta) \\ & -\gamma\mathrm{Atan}(\sigma e_\eta)^T e_f - \mathrm{Atan}(\xi)^T K_i K_f\mathrm{Atan}(\xi) \end{aligned} \tag{54}$$

Using (6) in Property 1, (9) in Property 4, (43), and (38), $\dot{V}_2$ is upper bounded by

$$\begin{aligned} \dot{V}_2 \le& -\rho\lambda_m(K_d)\|\dot{e}_\eta\|^2 + \rho\bar{e}_f\|\dot{e}_\eta\| + \gamma\sigma\lambda_M(M)\|\dot{e}_\eta\|^2 \\ & +\gamma C_M\|\dot{\eta}\|\|\mathrm{Atan}(\sigma e_\eta)\|\|\dot{e}_\eta\| - \gamma\lambda_m(K_p)\|\mathrm{Atan}(\sigma e_\eta)\|^2 \\ & +\gamma\bar{e}_f\|\mathrm{Atan}(\sigma e_\eta)\| - \lambda_m(K_iK_f)\|\mathrm{Atan}(\xi)\|^2 \end{aligned} \tag{55}$$

According to Assumption 1, (42), and the fact that $\|\dot{\eta}\| \le \|\dot{\eta}_d\| + \|\dot{e}_\eta\|$, the fourth term of (55) is upper bounded by

$$\begin{aligned} & \gamma C_M\|\dot{\eta}\|\|\mathrm{Atan}(\sigma e_\eta)\|\|\dot{e}_\eta\| \\ \le\ & \gamma C_M(\|\dot{\eta}_d\| + \|\dot{e}_\eta\|)\|\mathrm{Atan}(\sigma e_\eta)\|\|\dot{e}_\eta\| \\ \le\ & \gamma C_M\nu\|\mathrm{Atan}(\sigma e_\eta)\|\|\dot{e}_\eta\| + \frac{\sqrt{3}}{2}\pi\gamma C_M\|\dot{e}_\eta\|^2 \end{aligned} \tag{56}$$

Therefore,

$$\begin{aligned} \dot{V}_2 \le\ & -\rho\lambda_m(K_d)\|\dot{e}_\eta\|^2 + \rho\bar{e}_f\|\dot{e}_\eta\| + \gamma\sigma\lambda_M(M)\|\dot{e}_\eta\|^2 \\ & +\gamma C_M\nu\|\mathrm{Atan}(\sigma e_\eta)\|\|\dot{e}_\eta\| + \frac{\sqrt{3}}{2}\pi\gamma C_M\|\dot{e}_\eta\|^2 \\ & -\gamma\lambda_m(K_p)\|\mathrm{Atan}(\sigma e_\eta)\|^2 + \gamma\bar{e}_f\|\mathrm{Atan}(\sigma e_\eta)\| \\ & -\lambda_m(K_iK_f)\|\mathrm{Atan}(\xi)\|^2 \\ \le\ & -\zeta^T Q\zeta - \lambda_m(K_iK_f)\|\mathrm{Atan}(\xi)\|^2 \\ & +\max\{\gamma,\rho\}\bar{e}_f\|\zeta\| \end{aligned} \tag{57}$$

where

$$\zeta = \begin{bmatrix} \|\mathrm{Atan}(\sigma e_\eta)\| \\ \|\dot{e}_\eta\| \end{bmatrix}$$

and

$$Q = \begin{bmatrix} \gamma\lambda_m(K_p) & -\frac{\gamma}{2}C_M\nu \\ * & \rho\lambda_m(K_d) - \gamma\sigma\lambda_M(M) - \frac{\sqrt{3}}{2}\pi\gamma C_M \end{bmatrix}$$

To guarantee $\dot{V}_2 < 0$, Q should be positive definite, which is equivalent to

$$\sigma < \frac{\lambda_m(K_p)\left(\rho\lambda_m(K_d) - \frac{\sqrt{3}}{2}\pi\gamma C_M\right) - \frac{\gamma}{4}C_M^2\nu^2}{\gamma\lambda_m(K_p)\lambda_M(M)} \tag{58}$$

which is always satisfied if the controller is parameterized as (46).

Moreover, (57) also indicates that

$$\dot{V}_2 \leq -\lambda_m(Q)\|\zeta\|^2 - \lambda_m(K_iK_f)\|\mathrm{Atan}(\xi)\|^2 + \max\{\gamma,\rho\}\bar{e}_f\|\zeta\| \tag{59}$$

It follows that if

$$\|\zeta\| \geq \frac{\max\{\gamma,\rho\}}{\lambda_m(Q)}\bar{e}_f \triangleq \Delta_\zeta \tag{60}$$

then $\dot{V}_2 \leq 0$, which means V_2 is decreasing and bounded, and ζ enters a small neighborhood of the origin B_{Δ_ζ}. Even though ζ enters B_{Δ_ζ}, it may move in and out since the negativity of $\dot{V}_2$ in B_{Δ_ζ} is not guaranteed. However, when ζ leaves the neighborhood, $\dot{V}_2$ becomes negative again and drives it back to B_{Δ_ζ}. Therefore, ζ is guaranteed to be UUB. As a result, states e_η, $\dot{e}_\eta$, and ξ are also bounded in a small neighborhood of the origin, and the closed-loop system is UUB. The proof is completed. □

4. Simulation and Experimental Results

The experimental platform is a custom-built quadrotor based on the QAV-250 frame, featuring an axis-to-axis diagonal length of 250 mm. The airframe utilizes a carbon fiber-reinforced structure to ensure rigidity while minimizing inertia. There are four EMAX MT2204 KV2300 brushless motors paired with 5045 self-tightening propellers. Motor commands are executed via EMAX Simonk 12A electronic speed controllers (ESCs). Attitude feedback is provided by an ICM20602 IMU module, which integrates a 3-axis gyroscope and accelerometer. The ANO-Pioneer flight control board, equipped with an STM32F407VG microcontroller (168 MHz clock speed), runs a real-time operating system to achieve a 200 Hz control loop frequency. This configuration reflects typical small-scale quadrotor dynamics, ensuring the controller's applicability to similar UAV platforms.

4.1. Parameters

To begin with, some necessary parameters of the quadrotor are measured or estimated. The brushless DC motors are driven by given ESC signals, and the relation between the angular speed of each rotor ω_i $(i = 1,2,3,4)$ and PWM n_i signal can be experimentally determined by ${\omega_i}^2 = k_1 n_i + k_2$, where $k_1 = 8.1270 \times 10^6$ and $k_2 = -3.5866 \times 10^6$, as shown in Figure 2. For a desired ω_i designed by (5), the ESC signal n_i can be calculated. The rest of the parameters are $l_1 = 0.077$ m, $l_2 = 0.102$ m, $I_{xx} = 0.0026$ kg·m^2, $I_{yy} = 0.0027$ kg·m^2, $I_{zz} = 0.0047$ kg·m^2, $k_T = 1.468 \times 10^{-6}$ N·s^2/rad^2, $k_\tau = 1.446 \times 10^{-8}$ N·m·s^2/rad^2 [29].

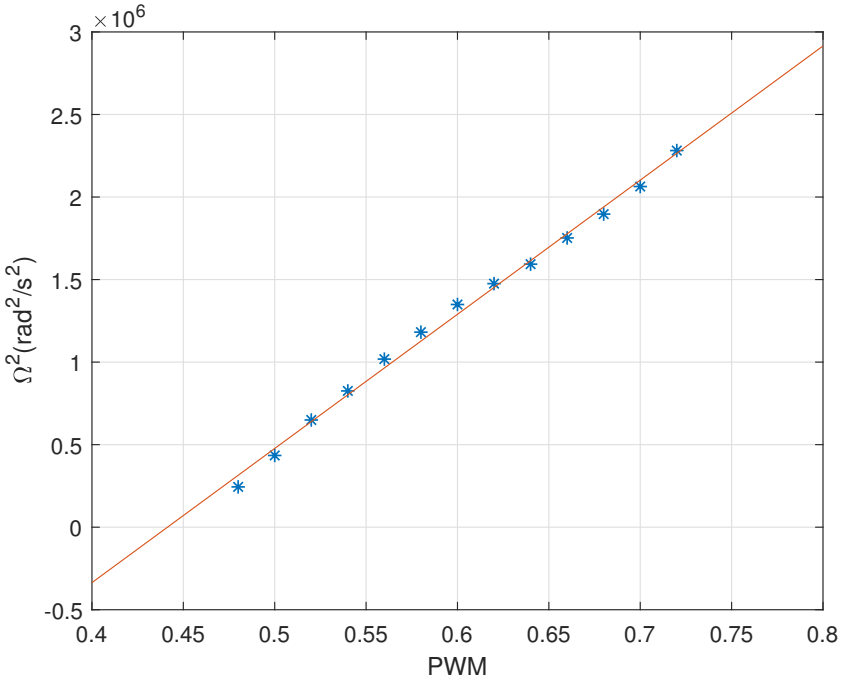


Figure 2. Relationship between ESC signal and the angular speed of rotor.

The control parameters are the same in subsequent simulations and experiments, which are as follows:

$$K_p = \text{diag}\{0.04, 0.04, 0.01\},\ K_i = \text{diag}\{0.01, 0.01, 0.002\},$$
$$K_d = \text{diag}\{0.02, 0.02, 0.005\},\ K_f = \text{diag}\{0.05, 0.05, 0.01\},$$
$$\sigma = 1.5, \gamma = 0.02, \rho = 0.02, \omega = 0.6, \beta = 0.75, \varepsilon = 0.05, \delta = \frac{1}{9}$$

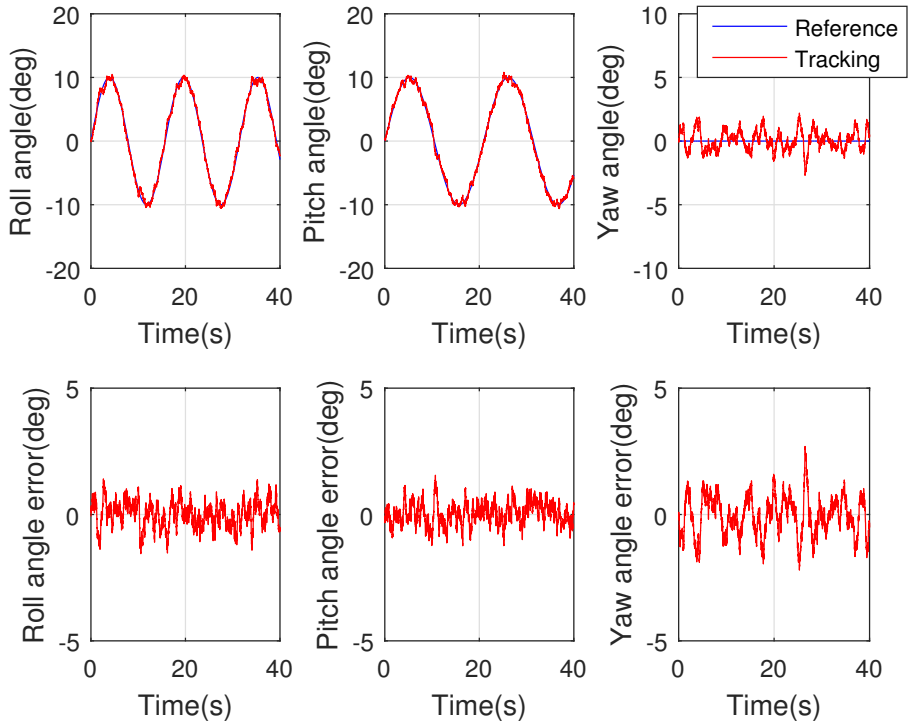
4.2. Simulation Results

In the simulation, the quadrotor attitude tracks a reference attitude command, similar to [24], $\eta_d = [10\sin(0.4t)\ 10\sin(0.3t)\ 0]^T$ and a set of sinusoidal wave disturbances is added as $\tau_d = 0.001[\sin(0.5t)\ \sin(0.5t)\ 0.1\sin(0.5t)]^T$ N·m. The initial conditions are $\eta(0) = [0,\ 0,\ 0]^T$ rad and $\dot{\eta}(0) = [0,\ 0,\ 0]^T$ rad/s. The initial values of z_1, z_2, and ξ are set to zeros. The simulation is carried out in MATLAB 2016a with a sampling time of 5 ms.

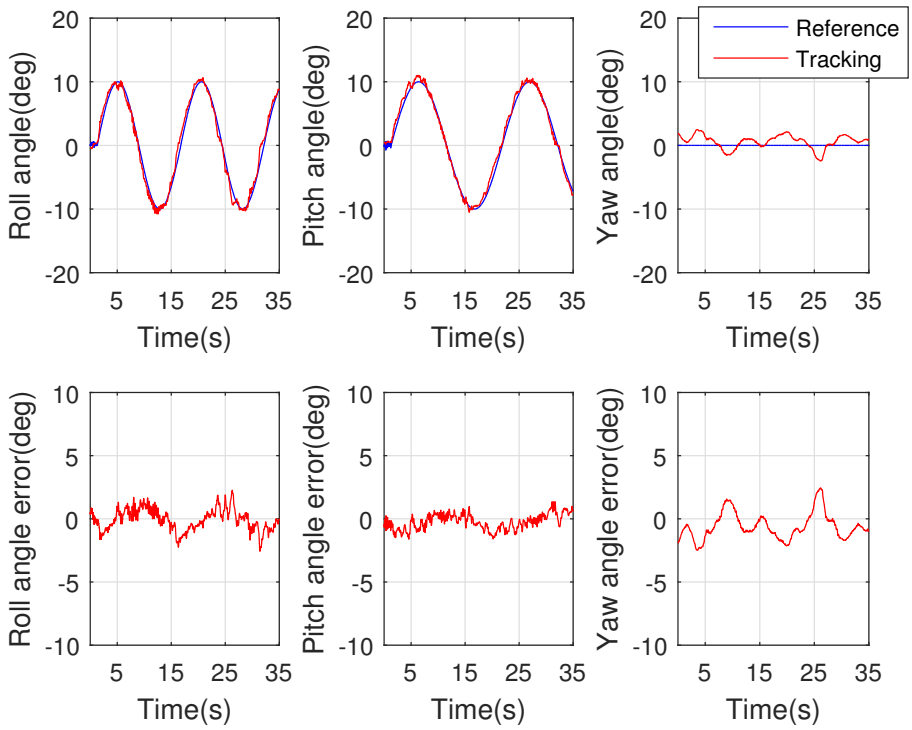
By utilizing the ESO, modeling error can also be considered as internal disturbance and can be compensated for. Here, the modeling error is assumed to be mainly caused by inaccurate measurement of moments of inertia, and a bias of 30% is added in simulations. In addition, considering the case in practical application, a measurement noise with zero mean and covariance of 0.01 and a system noise with zero mean and covariance of 10^{-6} are added in attitude angle and angular acceleration, respectively.

The simulation results are shown in Figures 3a, 4a and 5a. In Figure 3a, it can be observed that the attitude angles can track the reference smoothly in the presence of lumped disturbance and noises. In Figure 4a, the control torques for the roll, pitch, and yaw axes are depicted, generated by the proposed ESO-based nonlinear PID controller. These torques exhibit smooth variations despite the applied lumped disturbance (sinusoidal wave) and measurement/system noises, demonstrating the controller's ability to stabilize the system. The absence of abrupt spikes indicates effective disturbance rejection through feedforward compensation from the ESO. Figure 5a illustrates the ESO's estimation performance, where the estimated lumped disturbance (combining external torque, modeling errors, and noise)

closely tracks the actual disturbance profile. Minor discrepancies arise due to the 30% inertia bias and added noises in the simulation, yet the bounded estimation errors align with the theoretical convergence guarantees. Together, these figures validate the ESO's role in actively mitigating disturbances and the controller's robustness in maintaining precise attitude tracking under noisy conditions.

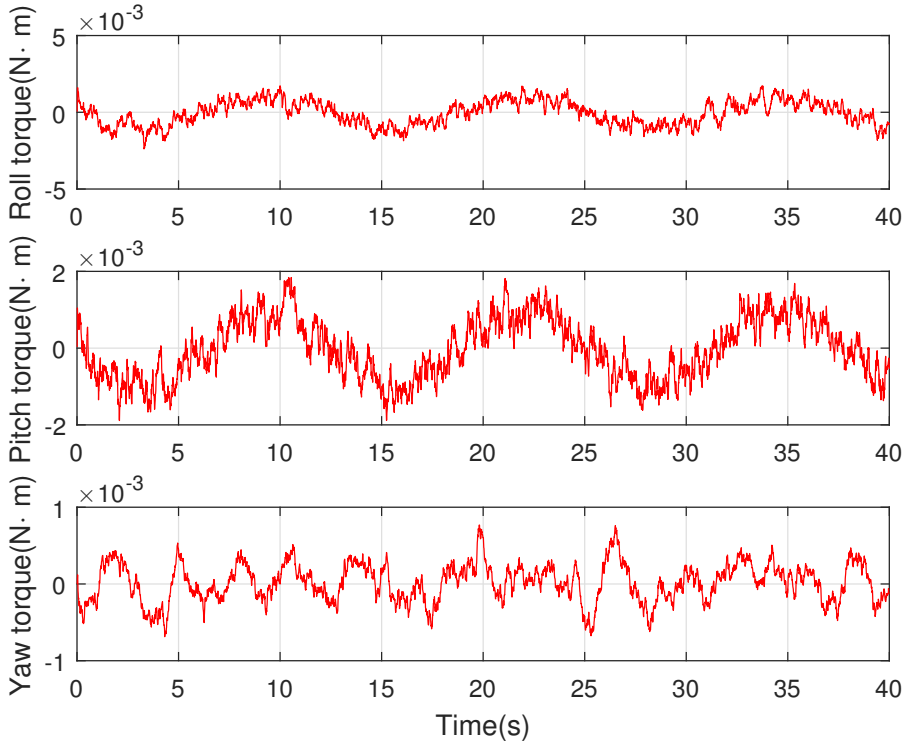


(a) Simulation results under lumped disturbance.

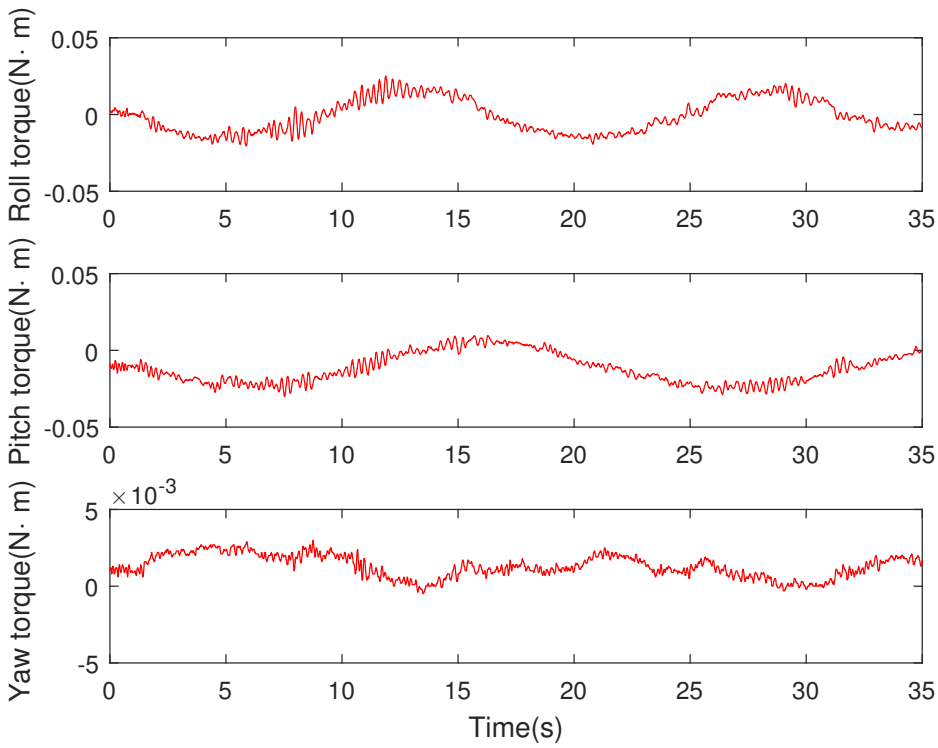


(b) Experimental results with measurement noise.

Figure 3. Attitude tracking performance and errors.

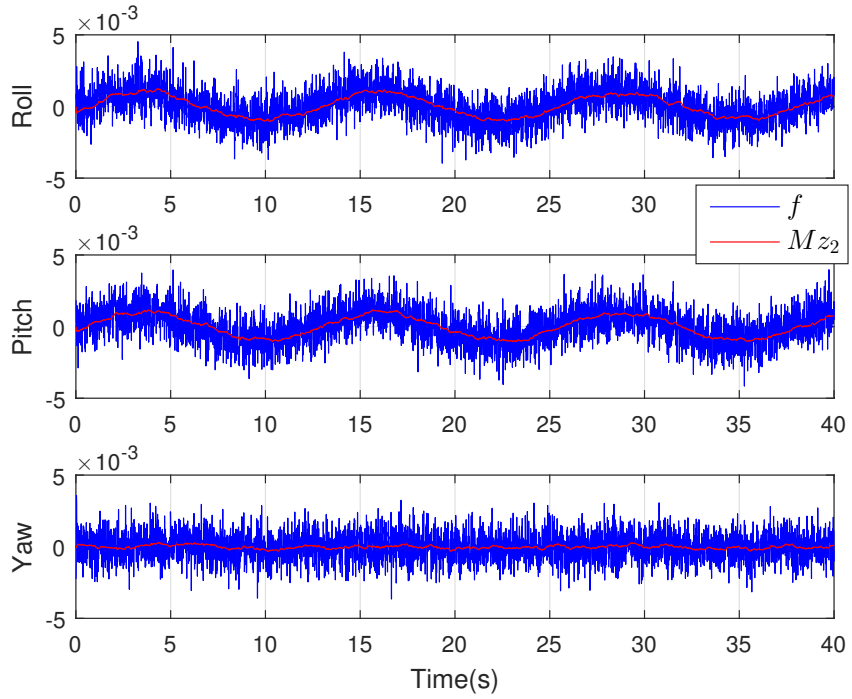


(a) Simulated torques.

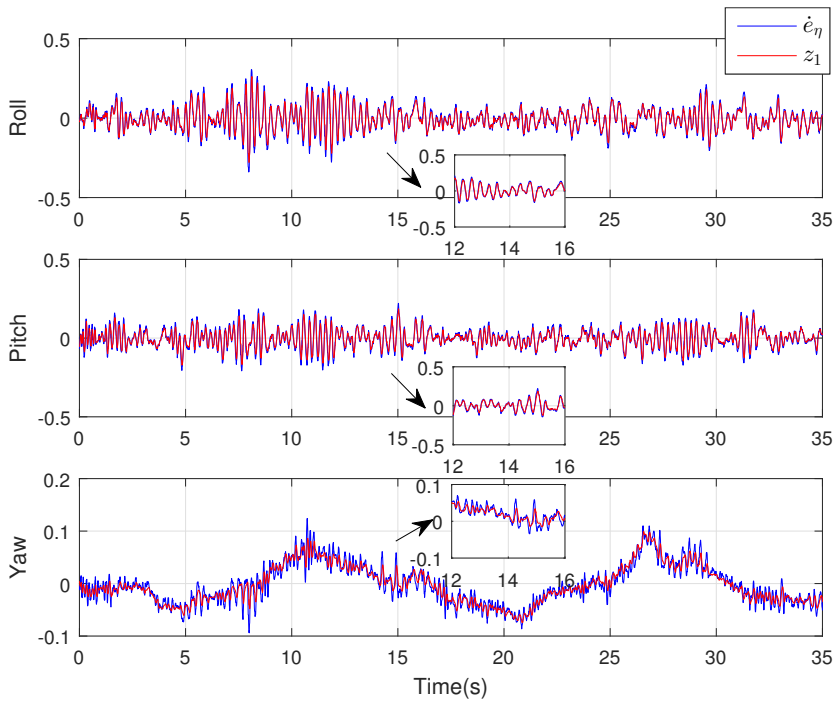


(b) Experimental torques.

Figure 4. Control torques of roll, pitch, and yaw axes.



(a) Simulation results.

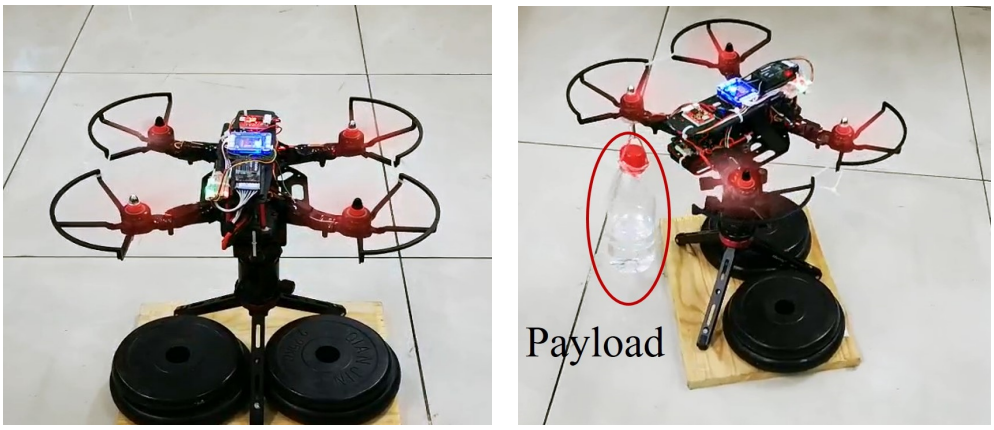


(b) Experimental results.

Figure 5. Disturbance estimations of proposed ESO.

4.3. Experimental Results

In this section, experiments are conducted to illustrate the effectiveness of the proposed method. To this end, an attitude experimental platform is built, as shown in Figure 6a. The quadrotor is fixed to a smooth joint and the translational movement is restricted. The hardware configuration is summarized in the beginning of this section. The attitude control frequency is 200 Hz and the updating frequency of the ESO is 1000 Hz.



(a) Experimental platform. (b) Experiment with payload.

Figure 6. Quadrotor attitude experimental platform.

4.3.1. Attitude Tracking Control

First, experiments are done to show the attitude tracking performance of the designed controller. The reference trajectory, initial values, and controller gains are set up identically to the simulations. In this case, the experimental results are depicted in Figures 3b, 4b and 5b. The oscillations in attitudes, control torques, and ESO estimations are due to the measurements noise of the IMU and the high-frequency dynamics caused by the vibration of quadrotor. Figure 3b shows the attitude tracking performance, including the reference commands and measurements of roll, pitch, and yaw angles. It was found that the tracking errors are driven into a small neighborhood of the origin, which agrees with the theoretical analysis. To show the efficiency of the ESO, the estimations of the derivatives of attitude tracking errors are also depicted in Figure 5b. The results indicate that the proposed reduced-order ESO can make a relatively accurate estimation even in the presence of measurement noises and high-frequency dynamics.

While experimental results include noise-induced oscillations (Figure 3b), the tracking errors remain bounded within theoretical predictions (Figure 3a). The ESO's disturbance estimates (Figure 5a vs. Figure 5b) exhibit consistent performance, with experimental deviations attributable to unmodeled high-frequency dynamics, yet still ensure effective compensation.

To quantitatively evaluate the controller's performance, key metrics such as maximum absolute error (MAE) and root mean square error (RMSE) are computed for both simulation and experimental results. For instance, in Figure 3a, the roll and pitch tracking errors exhibit MAEs of 0.12 rad and 0.11 rad, respectively, with RMSEs of 0.08 rad (roll) and 0.07 rad (pitch). The yaw channel shows superior precision, achieving an MAE of 0.018 rad and RMSE of 0.009 rad under disturbances. Experimental results (Figure 3b) reflect slightly higher but still bounded errors due to real-world noise and vibrations: MAEs of 0.16 rad (roll), 0.15 rad (pitch), and 0.025 rad (yaw), with RMSEs of 0.10 rad, 0.09 rad, and 0.015 rad, respectively. Control torque signals (Figure 4) remain within ± 0.25 N·m (roll/pitch) and ± 0.05 N·m (yaw), demonstrating smooth actuation. The ESO's disturbance estimation error

(Figure 5) maintains an RMSE of 0.003 N·m in simulations and 0.008 N·m in experiments, confirming its efficacy despite noise. These quantitative results align with the theoretical UUB guarantees and validate the algorithm's robustness.

Furthermore, in order to verify that ESO is likely to enhance the robustness of system, another set of experiments will be discussed in Section 4.3.2.

4.3.2. Attitude Stabilization with Disturbance Torque

In this paper, to illustrate the robustness of the proposed method, experiments were carried out in which a payload of 200 g was hung on one arm of the quadrotor at a time instant. It can be approximately considered that a time-varying disturbance torque acts on the quadrotor as the payload swings. The experimental platform is shown in Figure 6b. All the parameters are the same as the previous ones. The results are given in Figures 7–9, and the moment that disturbance torque joins is marked in the figures. Figure 7 indicates that there is no abrupt change in attitude angles and the tracking error is still acceptable even subjected to disturbance torque. It can also be observed from Figures 8 and 9 that the disturbance torque can be estimated quickly and compensated for actively in control torques. It is considered that the ESO plays a key role in disturbance rejection, and the proposed method demonstrates a good robustness and tracking accuracy.

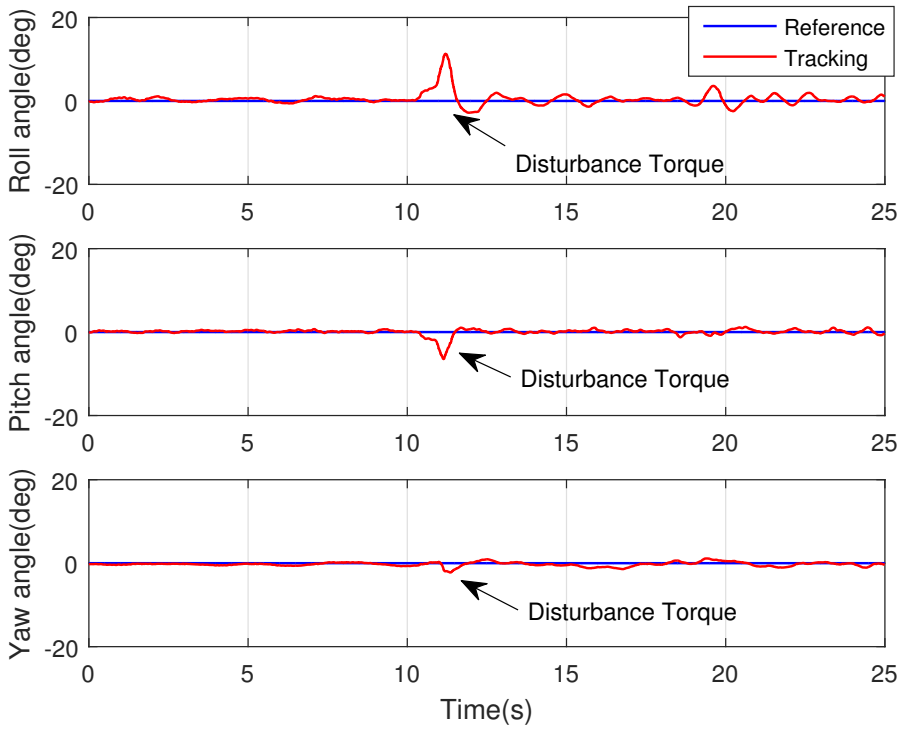


Figure 7. Experimental results: attitude tracking performance with payload disturbance.

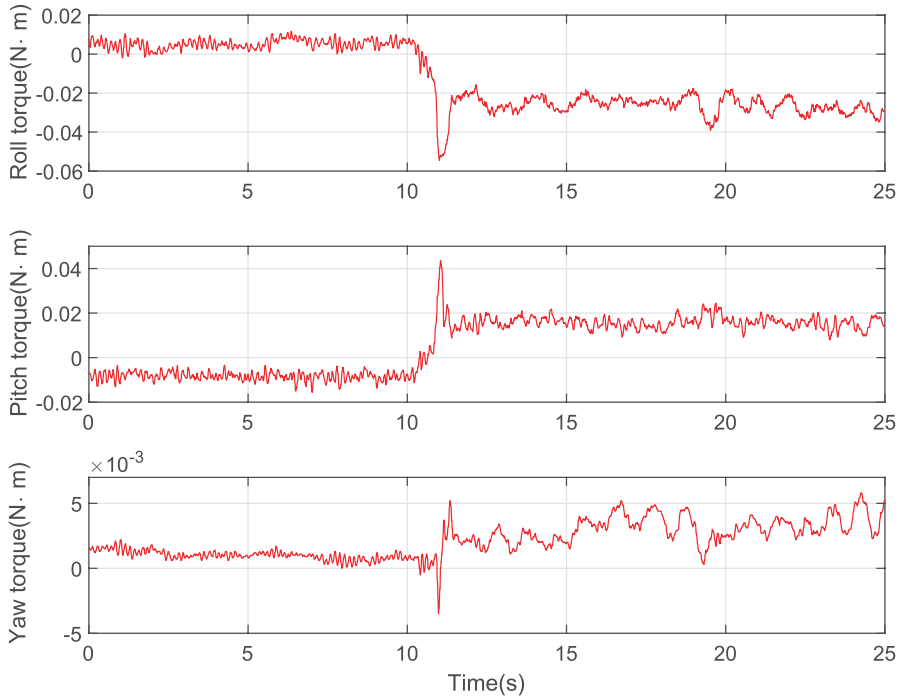


Figure 8. Experimental results: control torques for roll, pitch, and yaw angles in payload disturbance scenario.

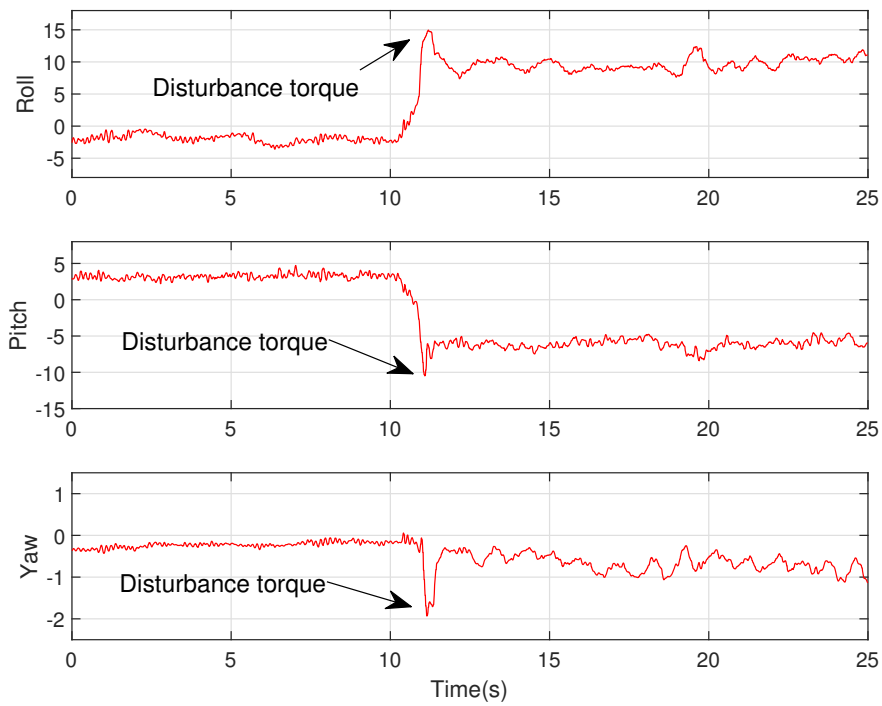


Figure 9. Experimental results: estimation performance of the proposed ESO in payload disturbance scenario.

5. Conclusions

In this paper, a nonlinear PID controller was investigated for the quadrotor attitude tracking problem. First, the attitude error dynamics were obtained with a lumped disturbance, and a special form of ESO was designed to estimate and compensate for the disturbance. Since the angular velocities were available via IMU, a reduced-order ESO

was applied to reduce the phase lag, as well as the influence of measurement noise and high-frequency dynamics. Then, an ESO-based nonlinear PID controller was designed for attitude tracking. Rigorous proof of the convergence of the closed-loop system in the presence of disturbances was provided based on the Lyapunov method. Finally, simulations and experimental results indicate a good tracking accuracy and robustness against disturbance of the proposed method. Despite the proposed method excelling in disturbance rejection and tracking precision, limitations such as manual parameter tuning and actuator saturation warrant further investigation. Therefore, future work will focus on intelligent parameter adaptation (e.g., reinforcement learning) and anti-saturation control of the quadrotor to broaden its applicability.

Author Contributions: Methodology, G.X.; Software, X.D.; Validation, G.X.; Investigation, Y.H.; Writing—original draft, G.X.; Writing—review & editing, S.L.; Supervision, Y.H.; Funding acquisition, S.L. and Y.H. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported in part by the National Natural Science Foundation of China under grant 62203013, Anhui Province University Collaborative Innovation Project under grant GXXT-2020-069, and the Scientific Research Foundation for Introduced Talent Scholars of Anhui Polytechnic University under grant 2022YQQ050.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

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